KaryoFlow: Few-Step Rectified-Flow Chromosome Detection With Localization-Quality Calibration

Author Name, *Member, IEEE*, and Co-Author Name, *Senior Member, IEEE*

***Abstract*—Automated karyotyping must detect about 46 densely distributed chromosomes from 24 morphologically similar classes. Small extent, touching or overlap, weak boundaries, and near-saturated class confidence make precise localization and reliable ranking the main obstacles to high-accuracy detection. We present *KaryoFlow*, which addresses these task-specific difficulties through low-dimensional rectified box transport, identity-consistent refinement of adjacent proposals, and final localization-quality calibrated ranking. On two public chromosome datasets with markedly different object-scale distributions, KaryoFlow exceeds the strongest evaluated detector by relative mAP margins of 1.8% and 0.2%, respectively. Combining few-step inference with cascade- head distillation provides up to $2.73\times$ faster inference than the RF+Heun reference, while distillation reduces mAP by only 0.0040 relative to its six-head teacher. These results establish a localization-aware generative approach and provide a new direction for high-precision chromosome detection.**

***Index Terms*—Rectified Flow, object detection, localization quality, diffusion models, medical image analysis, chromosome karyotyping**

I. Introduction

Chromosome karyotyping remains a labor-intensive step in cytogenetic analysis. Its detection stage is structurally unusual: a metaphase spread contains a nearly fixed set of about 46 chromosomes assigned to 24 classes, many instances occupy less than one percent of the image, and neighboring chromosomes frequently overlap in their bounding-box support. In our two training cohorts, the median instance count is 46; 98.8% and 100% of images contain at least one pair of ground-truth boxes with IoU above 0.1000. The median relative box area also shifts from 0.4% to 0.8% between cohorts. Consequently, a few-pixel error can preserve AP50 while changing the strict-IoU ordering that determines mAP, and an iterative detector must refine many adjacent candidates without confusing their identities.

Conventional detectors offer complementary but incomplete solutions. Anchor-based systems such as Cascade R-CNN [1] and YOLOX [2] are fast, whereas transformer detectors such as DINO [3] obtain strong localization through multi-scale deformable attention [4]. DiffusionDet [5] instead treats boxes as a set generated from noise, an appealing formulation for variable spatial layouts, but its DDPM inheritance introduces a curved stochastic path and a many-step design that are poorly matched to interactive microscopy analysis. More importantly, simply shortening that path does not resolve two detection-specific mismatches: multistep integration assumes a persistent state identity, whereas proposal renewal replaces selected states; and final ranking by class confidence ignores whether a small chromosome is localized accurately enough to survive strict IoU thresholds.

Rectified Flow [6] (RF) offers a natural generation foundation by replacing the DDPM path with a deterministic linear transport between noise boxes and targets. Applying RF to

detection raises three linked questions: how to learn a well-conditioned box trajectory from limited data, how to integrate it without violating proposal identity, and how to rank fixed outputs by both semantic and geometric reliability. KaryoFlow answers them as one trajectory–decision design, shown in Figure 1. Its contributions are threefold.

First, we formulate RF directly in normalized four-dimensional box space and derive a data-prediction objective for the reverse path. Although data and velocity prediction are algebraically equivalent, their squared losses differ by t^{-2}; direct box prediction therefore avoids amplifying errors near the terminal state emphasized by the shifted schedule. The resulting direction is replicated across three seeds on both chromosome datasets. RF itself is adopted, and our claim is the detection formulation and analysis rather than invention of the underlying flow framework.

Second, we derive an RF data-prediction update from DPM-Solver++ [7] and establish the proposal-identity condition for multistep box integration. If a row is renewed between two evaluations, the history correction becomes a secant between different candidates and is not a derivative along either trajectory. Disabling renewal when the proposal pool is redundant restores this premise, retains three-seed accuracy on both datasets, and saves 2.43% latency; its failure at $K = 100$ provides an explicit capacity boundary. DPM-Solver++ itself is used for efficiency, not claimed as an accuracy novelty.

Third, we introduce Localization-Quality Calibrated Ranking (LQCR) for the strict-IoU bottleneck of small, crowded chromosomes. From the final proposal feature, LQCR predicts conditional localization quality and ranks detections by $s = pq^2$. It is applied only when outputs are emitted, leaving candidate coordinates, class logits, proposal survival, and the entire RF trajectory unchanged before final post-processing. The resulting fixed-checkpoint gain is therefore attributable to decision-stage ranking.

We evaluate KaryoFlow on two public chromosome datasets

Manuscript received [date]; revised [date]. (Corresponding author: Author Name.)

The authors are with [Affiliation], [City], [Country] (e-mail: author@example.com).

containing $2{,}200$ and $5{,}000$ images, with multi-seed controls, same-checkpoint solver tests, test-set evaluation, strict-IoU analysis, and same-hardware latency profiling. The experiments also separate solver network evaluations from cascade-head evaluations and show that head distillation and conditional stepping compress orthogonal sources of inference cost. Standardized coupling controls are null, so stochastic coupling is excluded from the final model and contribution list.

II. Related Work

a) Chromosome detection and karyotyping: A metaphase image combines several difficulties rarely present together in natural-image detection: dozens of adjacent objects, homologous classes with similar morphology, scale variation between chromosome groups, staining shift, and incomplete boundaries caused by touching or overlap. Earlier work progressed from joint segmentation and classification [8], [9] to learned enumeration, straightening, and metaphase detection [10], [11], [12]. Recent 2026 studies confirm that the clinical workflow is moving toward direct detection from whole metaphase images: object detectors remove manual single-chromosome cropping [13], and integrated pipelines combine YOLO-based detection with pairing and anomaly screening [14]. However, those reports emphasize concordance, mAP50, or downstream workflow accuracy. These measures can remain high when a small chromosome has a several-pixel boundary error. Transfer-learning failure analysis likewise identifies overlap-related cropping artifacts as a persistent source of chromosome confusion [15]. KaryoFlow therefore targets the localization stage itself, with mAP over IoU $0.50{:}0.95$, strict-IoU diagnostics, and measured inference cost.

Large-scale representation learning addresses a complementary limitation. CHROMA was pretrained on more than four million chromosome images and improves robustness to rare abnormalities and challenging acquisition conditions [16]. Such single-chromosome representations reduce annotation burden after extraction, but do not resolve how densely overlapping candidates should move, retain identity, or be ranked in the preceding box-detection stage. This distinction motivates a detector specialized for metaphase box geometry rather than another isolated-chromosome classifier.

b) Generative detection and few-step integration: DiffusionDet [5] formulates object detection as iterative box denoising. FlowDet [17] and DeFloMat [18] replace DDPM-style dynamics with conditional flow matching or RF, while DiffuBox [19] studies a distinct 3D refinement setting. RF [6] and Flow Matching [20] provide our adopted continuous-time foundation, and DPM-Solver++ [7] supplies the multistep integration principle. Recent work has also explored flow-matching design and optimized priors for cellular-microscopy generation and medical augmentation [21], [22]; these image-generative settings are complementary to our discriminative transport of box sets. Existing studies emphasize the probability path or solver, but do not analyze the interaction between solver history and proposal renewal. KaryoFlow derives the data-prediction update for the RF box path and makes proposal identity an explicit condition of multistep inference. Diffusion in TMI has mainly addressed synthesis, segmentation, or robust classification [23], [24], [25], leaving few-step medical box-set prediction comparatively underexplored.

c) Localization quality in medical detection: Medical detectors must handle small targets, incomplete labels, acquisition shift, and structured false positives. Representative TMI studies address few-shot pulmonary lesions, heterogeneous lesion annotations, and relational transfer across domains [26], [27], [28], while context refinement, weak supervision, and imbalanced learning reduce task-specific false positives [29], [30], [31]. For general detection, IoU-Net predicts localization confidence for proposal selection [32], and GFL and Varifocal-Net combine class and localization quality in dense heads [33], [34]. LQCR inherits the probability-ranking motivation but differs in causal scope: it is applied only after a generative detector has completed all trajectory updates, so solver history, coordinates, predicted classes, and proposal survival are fixed. This controlled boundary is particularly useful in chromosome images, where AP50 is nearly saturated but strict-IoU errors remain concentrated in small and overlapping instances. The emphasis on output reliability is also consistent with evidence that accurate medical networks can remain poorly calibrated [35].

III. Method

A. Box-Space Rectified Flow

a) Linear transport in chromosome-box space: Each chromosome is represented by a normalized center and size vector (c_x, c_y, w, h). Training connects a Gaussian source box $\mathbf{x}_1$ to a matched chromosome box $\mathbf{x}_0$ through

$$\mathbf{x}_t = (1-t)\,\mathbf{x}_0 + t\,\mathbf{x}_1, \qquad t \in [0,1], \tag{1}$$

whose target velocity is $\mathbf{x}_1 - \mathbf{x}_0$. Unlike image diffusion, only four box coordinates are transported; the microscopy features remain fixed conditioning information. This low-dimensional, straight path is a natural match for a metaphase spread, where approximately 46 boxes must move concurrently while preserving their individual identities.

b) Time conditioning: AdaLN-Zero [36] injects the time embedding into each proposal head through zero-initialized residual modulation. It is retained as a stable conditioning mechanism; controlled ablation does not support treating it as a separate accuracy contribution.

B. Stable Parameterization and Few-Step Integration

a) Few-step integration: Euler integration provides the first-order reference: it is inexpensive per step but uses only the current box estimate. Heun adds a predictor–corrector evaluation to reduce local integration error, so four reverse steps require seven network evaluations. DPM-Solver++ [7] instead reaches second order by reusing the previous target-box prediction, requiring only one new evaluation per step. We compare the solvers in this progression—Euler, Heun, then DPM-Solver++—and attribute solver gains only under matched checkpoints and step counts.

The model directly predicts $\mathbf{x}_0$ (the target box), and the update uses polynomial interpolation of the $\mathbf{x}_0$ history in t

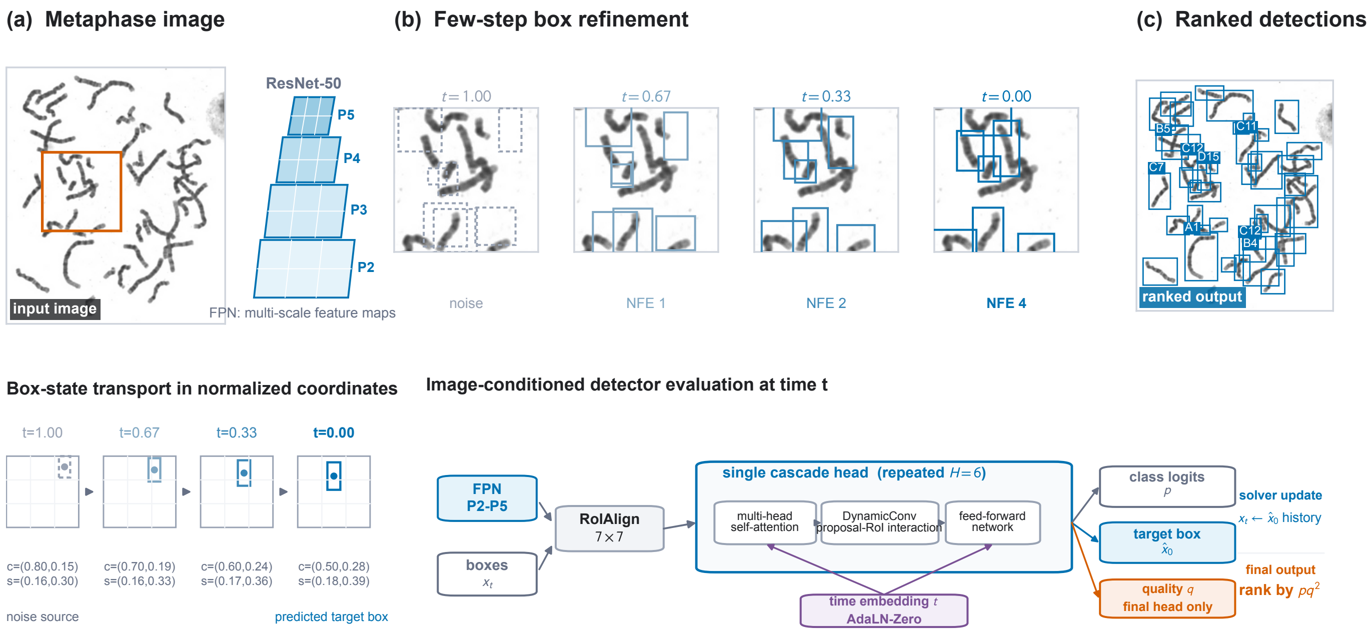

Fig. 1. KaryoFlow from image to detections. A real metaphase image and its dense crop show four-NFE box refinement and the final ranked detections. The lower row makes one normalized center-and-scale trajectory explicit and exposes an actual detector evaluation: multi-scale RoIAlign, self-attention, DynamicConv, feed-forward refinement, AdaLN-Zero time conditioning, six cascade heads, and the final-head quality branch. Endpoints and output boxes are empirical; intermediate boxes visualize the prescribed linear RF path rather than recorded hidden activations. Vector overlays remain editable while microscopy panels retain the source pixels.

space (not the log-SNR λ space of VP-SDE). The RF-ODE in data-prediction form is $d\mathbf{x}/dt = (\mathbf{x} - \mathbf{x}_0(t))/t$; integrating exactly over $[t_n, t_{n+1}]$ with linear $\mathbf{x}_0(t)$ interpolation yields the update in (2)

$$\begin{aligned} \mathbf{x}_{t_{n+1}} &= \tfrac{t_{n+1}}{t_n}\,\mathbf{x}_{t_n} + \left(1 - \tfrac{t_{n+1}}{t_n}\right)\mathbf{x}_0^{(n)} + \varphi_1\,\mathbf{D}_1, \\ \varphi_1 &= t_{n+1}\log\tfrac{t_n}{t_{n+1}} - t_n + t_{n+1},\ \mathbf{D}_1 = \tfrac{\mathbf{x}_0^{(n)} - \mathbf{x}_0^{(n-1)}}{t_n - t_{n-1}}, \end{aligned} \tag{2}$$

The first two terms move a proposal toward its current box estimate; the last term corrects that move using the change between the two most recent estimates. The initial noise sample remains fixed. Thus four steps require four rather than seven network evaluations. The endpoint is protected with $t_{n+1} > 10^{-7}$.

b) Why direct data prediction is numerically preferable here: For the path in (1), $\mathbf{x}_0 = \mathbf{x}_t - t\mathbf{v}$. Hence the two parameterizations carry the same information, but their pointwise squared errors satisfy

$$\mathcal{L}_v(t) = t^{-2}\mathcal{L}_{x_0}(t). \tag{3}$$

With the shifted schedule, velocity regression therefore amplifies gradient noise as $t \to 0$, where accurate terminal boxes matter most. Direct x_0 prediction removes this avoidable conditioning factor. The predicted $+0.0020$ validation advantage is reproduced by three seeds on each dataset; test differences remain within noise, so the result supports stable optimization rather than universal statistical dominance.

c) Euler, Heun, and DPM-Solver++ comparison: Evaluating the RF+Heun checkpoint across all solver×step combinations (Table I), solver type has *no effect on* mAP at matched steps (Euler = DPM-Solver++ at both 4-step and 1-step). Step count has marginal effect ($+0.0040$ from 1 to 4 steps). Heun's

TABLE I
SOLVER×STEP DISENTANGLEMENT ABLATION ON THE RF+HEUN CHECKPOINT (DATASET 2 VAL, SEED 42). SOLVER TYPE IS MAP-NEUTRAL AT MATCHED STEPS, WHILE INCREASING ONE TO FOUR STEPS ADDS ONLY 0.0040 MAP.

Solver	Steps	NFE	mAP
Euler	1	1	0.8510
Euler	4	4	0.8550
Heun	4	7	0.8560
DPM-Solver++	1	1	0.8510
DPM-Solver++	4	4	0.8550

$+0.0010$ over Euler 4-step costs $1.75\times$ NFE (7 vs 4), not cost-effective. A second same-checkpoint check on the RF-DPM++ model gives 0.8630 for four-step DPM-Solver++ and 0.8640 for four-step Heun ($\Delta = -0.0010$). The solver is therefore an efficiency component rather than an accuracy component.

d) Conclusion: DPM-Solver++ is an efficiency component. On the same RF-DPM++ checkpoint at four steps, DPM-Solver++ and Heun differ by only -0.0010 aggregate mAP, while DPM-Solver++ uses four rather than seven network evaluations. The previously observed $+0.0060$ cross-checkpoint difference compared models trained under different configurations and cannot be attributed to the inference solver alone. We therefore claim matched accuracy at lower NFE, not a solver-induced accuracy improvement.

C. History-Consistent Proposal Integration

a) Proposal identity is a premise of multistep history: Let $r_i^n = 1$ indicate that proposal row i is renewed after step n.

Proposition 1 (Identity consistency)**.** *The multistep correction $\mathbf{D}_{1,i}$ in* (2) *is a consistent first-order approximation to a trajectory derivative only if the two states indexed by row i belong to the same proposal trajectory, or if the row-wise history is reset after replacement.*

Proof. Without renewal, differentiability of $\mathbf{x}_{0,i}(t)$ gives the standard secant limit

$$\mathbf{D}_{1,i} = \frac{\mathbf{x}_{0,i}^{(n)} - \mathbf{x}_{0,i}^{(n-1)}}{t_n - t_{n-1}} \longrightarrow \frac{d\mathbf{x}_{0,i}}{dt}. \tag{4}$$

If $r_i^n = 1$, the numerator subtracts predictions from two independently initialized states. Its limit contains a finite cross-trajectory jump divided by $t_n - t_{n-1}$ and therefore is not the derivative along either trajectory. Resetting history removes the invalid correction. □

We use the simpler alternative of disabling renewal after verifying sufficient proposal redundancy. This choice is well matched to metaphase images: the object count is tightly concentrated near 46, so $K = 500$ provides multiple candidates per chromosome, while maintaining identity avoids mixing adjacent, morphologically similar instances. The $K = 100$ counterexample, where mAP drops by 0.0313, shows that history consistency cannot compensate for inadequate proposal capacity.

D. Orthogonal Inference Compression

a) Cascade-head distillation: Solver NFE counts network calls, whereas each call evaluates H cascade heads. We distill a six-head teacher into a three-head student using head mapping $\{0\to 0, 1\to 2, 2\to 5\}$ and feature/prediction supervision. At four solver steps this reduces cascade-head evaluations (CHE) from 24 to 12, while NFE remains four; the reduction is therefore 24-to-12 CHE, not NFE. The compressed model reaches 0.8590 versus 0.8630 and reduces RTX A6000 latency from 75.20 to 44.10 ms. Together with few-step integration, this reduces latency from the 120.34-ms RF+Heun reference to 44.10 ms, a 2.73× acceleration; the 0.0040 mAP cost is measured against the six-head teacher.

b) Conditional computation: Geometry-aware adaptive conditional stepping (GACS) freezes convergence and confidence thresholds on seed 42, then chooses one or two steps per image. It is an optional speed policy rather than the accuracy model: on Dataset 1 it uses 1.6136 mean steps, exits early on 38.6% of images, and reduces latency by 13.8% at −0.0007 mAP versus fixed two-step inference. Its mAP remains below the four-step reference, so we do not call it lossless relative to the production setting.

E. Localization-Quality Calibrated Ranking

Chromosome classes can be predicted confidently even when their boxes are a few pixels too loose or shifted. This is especially harmful for small or touching chromosomes because a small displacement causes a large IoU change. For proposal feature F, correct-class event C, IoU U, and evaluation threshold τ, the true-positive probability separates as

$$P(C{=}1, U{\geq}\tau \mid F) = P(C{=}1 \mid F)P(U{\geq}\tau \mid C{=}1, F). \tag{5}$$

The second factor is absent from class-only ranking. LQCR estimates it with a two-layer quality branch $q = \sigma(g_\phi(F))$ attached only to the final cascade head. It adds 33,153 parameters. The backbone, FPN, proposal interaction, class head, and box head are frozen during branch fitting, so the experiment isolates output calibration from box generation.

For Hungarian-matched proposal i, let $m_i = 1$ and let u_i be the detached aligned IoU; unmatched proposals have $m_i = 0$. We train the scalar logit with the continuous target $y_i = m_i u_i$ and weighted binary cross-entropy

$$\begin{aligned} \mathcal{L}_q &= \frac{\lambda_q}{\max(N_+, 1)} \sum_i w_i\, \ell_{\mathrm{BCE}}(g_i, y_i), \\ w_i &= m_i u_i + (1 - m_i)\alpha q_i^\gamma, \end{aligned} \tag{6}$$

where $\lambda_q = 0.2500$, $\alpha = 0.7500$, and $\gamma = 2$. Detaching u_i prevents calibration from changing box geometry. Matched proposals learn their IoU; unmatched proposals learn zero. At output, class probability p is replaced by $s = pq^2$. A validation sweep over $\beta \in \{0, 0.2500, 0.5000, 1, 2\}$ selects the exponent 2; this learned-quality sweep is separate from the true-IoU oracle used only to quantify headroom.

Our strict final-only implementation keeps raw class scores in the solver, proposal renewal, Top-K pruning, and early stopping. Calibration is applied only when emitting detections, so boxes, predicted classes, and RF trajectories remain fixed at the candidate level; score-dependent NMS or max-detection selection may consequently choose a different final subset. An elementwise checkpoint audit confirms that all 590 tensors shared with the six-head detector are identical; the LQCR checkpoint contains only five additional quality-branch tensors. This design targets the observed chromosome regime in which AP50 is nearly saturated but small and overlapping instances lose recall at IoU 0.9000. Figure 2 summarizes the principle, causal boundary, and fixed-seed controlled effect.

F. Coupling Audit

We evaluated Random, Hard-OT, and Sinkhorn-sampled coupling under the same augmentation and evaluation protocol. Although deterministic assignment can reduce pairing diversity in the four-dimensional box space, the controlled multi-seed experiments do not show a reliable mAP advantage for stochastic coupling on either chromosome dataset. This audit is motivated by conditional and multisample flow matching [37], [38] and uses entropic Sinkhorn transport [39]; it also agrees with evidence that conditional OT can introduce its own failure modes [40]. Coupling is therefore treated as an auxiliary design choice and negative-result analysis, not as a component of the final method or an accuracy contribution. The complete model uses Random coupling to remove this confound.

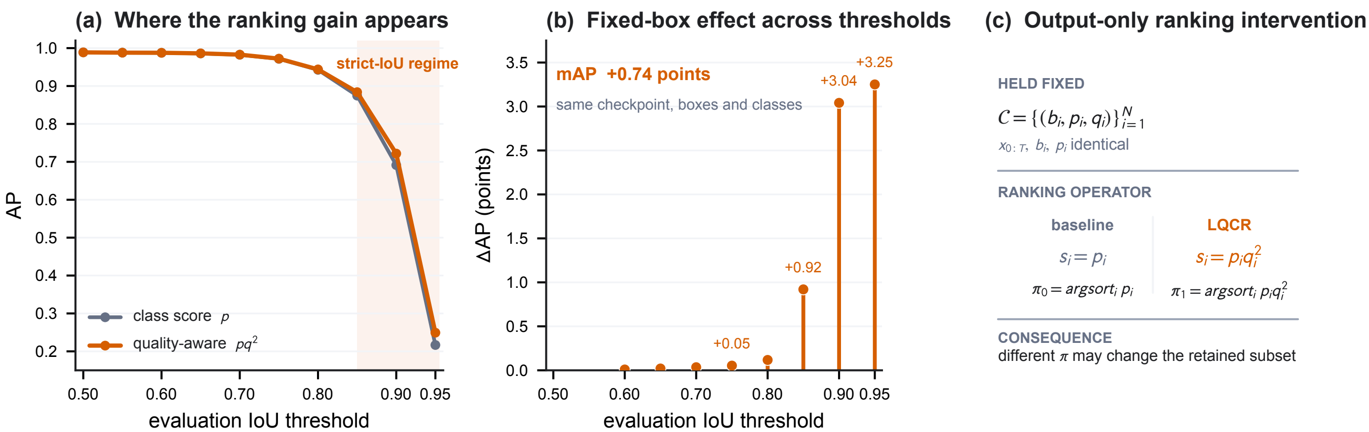

Fig. 2. Effect and causal scope of localization-quality ranking. (a,b) On the fixed Dataset 2 seed-42 RF-DPM++ checkpoint, the gain is concentrated at strict IoU thresholds (AP90/AP95 $+3.04/+3.25$ points), while AP50 is essentially unchanged (-0.01 points). (c) The intervention changes only final ordering from p to pq^2; the RF trajectory, renewal, candidate coordinates, and class logits are fixed. Score-dependent post-processing may select a different final subset. Values are loaded from checkpoint-linked exports.

TABLE II
DATASETS USED IN THIS PAPER. DATASET 1 IS CHROMOSOME20240904 (RST); DATASET 2 IS THE 24 CHROMOSOMES OBJECT DATASET.

Dataset	Train	Val	Test	Classes
Dataset 1	1,540	440	220	24
Dataset 2	3,500	500	1,000	24

G. Top-K Proposal Pruning

After step 0 of inference, we prune proposals from 500 to K based on confidence scores; only the top-K proposals proceed through steps 1–3. Because pruning changes the proposal set, multistep history is reset before the retained proposals continue refinement.

IV. EXPERIMENTS

A. Experimental Setup

a) Datasets: Table II summarizes the two public chromosome datasets, referred to as Dataset 1 and Dataset 2. Dataset 1 is Chromosome20240904 (RST) [41], with 2,200 images; Dataset 2 is the 24 Chromosomes Object dataset [42], [43], with $5{,}000$ images. Their training splits contain $1{,}540$ and $3{,}500$ images, respectively. Both have a median of 46 annotated instances per image, but their median relative box areas differ substantially (0.4% versus 0.8%). Under the COCO absolute-area definition, 23.0% of Dataset 1 instances are small, compared with only 0.4% in Dataset 2; correspondingly, 99.8% and 11.2% of training images contain at least one small instance. Moreover, 98.8% and 100.0% of training images contain at least one ground-truth box pair with IoU above 0.1000. These statistics quantify the small, crowded regime and the scale shift used for cross-dataset replication. Both datasets use image-level random splitting. A single blood sample may yield multiple metaphase images, but patient identifiers are unavailable; thus patient-level separation cannot be verified. We analyze only the publicly released images and annotations and collect no additional human-subject data.

TABLE III
TRAINING-SET PROPERTIES THAT MOTIVATE THE DETECTOR DESIGN. RELATIVE AREA IS BOX AREA DIVIDED BY IMAGE AREA; SMALL FOLLOWS THE COCO ABSOLUTE-AREA THRESHOLD ($< 32^2$ PIXELS); OVERLAP DENOTES AN IMAGE CONTAINING AT LEAST ONE PAIR OF GROUND-TRUTH BOXES WITH IOU > 0.1000.

Dataset	Median n	Med. rel. area	Small inst.	Images w/ small	Overlap
Dataset 1	46	0.4%	23.0%	99.8%	98.8%
Dataset 2	46	0.8%	0.4%	11.2%	100.0%

b) Architecture and training: Our base framework (KaryoFlow) uses ResNet-50 [44] + FPN [45] (256 channels, 4 levels), 500 proposals, 6 cascade transformer heads with deep supervision (5 auxiliary heads). Optimization: AdamW [46] (lr$=$ $5{\times}10^{-5}$, wd$=$ 10^{-4}), 5-epoch warmup + CosineAnnealing [47], 150 epochs. Loss: Focal [48] (λ_{cls}=2.0) + L1 (λ=5.0) + GIoU [49] (λ=2.0) with Hungarian matching [50]. Diffusion uses RF with shifted noise schedule (shift$=$ 3.0); default solver is DPM-Solver++ 4-step. For the isolated LQCR intervention, the trained six-head checkpoint initializes the model, all pre-existing parameters are frozen, and only the 33k-parameter quality branch is optimized for at most 12 epochs with AdamW learning rate 10^{-3} and cosine decay; the epoch-2 validation checkpoint is used. Thus this phase fits a ranking statistic but cannot alter the detector's generated boxes or class logits.

c) Detection protocol: Boxes are represented in two spaces: image space (absolute pixel xyxy) for RoIAlign and NMS, and diffusion space (cxcywh normalized to $[0, 1]$, mapped to $[-s, +s]$, s=2.0). The RF forward path is $x_t = (1-t)\, x_0 + t\, \varepsilon$ (DDPM baseline uses cosine schedule). At inference, 500 random-noise proposals are iteratively denoised; with time-ensemble, predictions from all steps are concatenated and deduplicated by per-class NMS (IoU 0.5000). Evaluation uses COCO mAP@0.5000:0.9500 (10 IoU thresholds, maxDets=100).

d) Statistical considerations: Cross-seed experiments (3 seeds: 42, 123, 789) are available for Dataset 1. For Dataset 2, the DPM-Solver++ variant has 3 seeds (mean 0.8590 ± 0.0030), which sets the scale for interpreting small differences. The $+0.0060$ cross-checkpoint solver difference compares independently trained models and is not a solver-causal result; the same-checkpoint comparison is neutral. LQCR has clean three-seed replication on Dataset 1 and a fixed-checkpoint, final-only intervention on Dataset 2. The latter isolates ranking from training and trajectory changes; the former tests whether the effect reproduces across independent training runs.

e) Reproducibility: All reported comparisons retain their dataset split, random seed, checkpoint, and evaluation protocol. Controlled results are separated from exploratory runs and diagnostic oracles. For LQCR, we additionally verify that all 590 shared detector tensors are identical before fitting the quality branch and distinguish learned-quality sweeps from true-IoU oracle analyses. This protocol prevents cross-checkpoint or oracle quantities from entering model claims.

B. Precision Bottleneck Diagnosis

Figure 3 separates classification, ranking, and coordinate headroom before proposing further modules. Perfect-class oracles add little compared with perfect-coordinate oracles, and AP degrades rapidly at strict IoU thresholds. At IoU 0.9000, small and overlapping chromosomes show the largest recall deficits. These observations identify localization accuracy and localization-aware ordering—rather than class recognition alone—as the current precision bottleneck.

C. Main Results: RF vs DDPM

a) Dataset 2 ablation: Table IV reports a cumulative ablation on the Dataset 2 validation set. The historical DDPM and RF+Heun rows establish the adopted generation foundation. The same seed-42 RF-DPM++ checkpoint then isolates the four-NFE DPM-Solver++ operating point and the final-only LQCR intervention. AdaLN-Zero is used throughout as time conditioning but is not counted as an independent contribution.

RF supplies the dominant historical improvement over the DDPM baseline. DPM-Solver++ retains accuracy while reducing Heun's seven evaluations to four; same-checkpoint solver tests do not support an additional precision claim. LQCR then improves mAP by 0.0074 without changing the checkpoint, candidate coordinates, class logits, renewal decisions, or trajectory. Its largest gains occur at strict IoU thresholds (AP90/AP95 $+0.0304/+0.0325$), consistent with the localization-aware ranking mechanism rather than a coordinate change.

Renewal, parameterization, and coupling conclusions use multi-seed controls; LQCR uses a fixed checkpoint and final-only causal boundary on Dataset 2 plus three-seed replication on Dataset 1; and GACS freezes its thresholds before cross-seed evaluation. Null directions are retained in the ablations rather than presented as model contributions.

b) Dataset 1: RF vs DDPM: On Dataset 1 (3 seeds), RF outperforms DDPM by $+0.0170$ mAP (0.7460 vs 0.7290, lower variance ± 0.0010 vs ± 0.0040) with 4-step inference vs DDPM's 1-step. DDPM gains only $+0.0440$ from $1 \rightarrow 8$ steps ($0.6280 \rightarrow 0.6720$), whereas the RF paradigm yields $+0.0820$ mAP on Dataset 2.

D. SOTA Comparison

Table VI uses one validation protocol for both chromosome datasets. The cohorts differ strongly in relative object scale, yet the full KaryoFlow model with LQCR obtains the highest mAP among the evaluated methods on each: 0.7550 on Dataset 1 and 0.8704 on Dataset 2. The margins over the strongest listed alternatives are 0.0130 and 0.0014, respectively. The result is not caused by a larger backbone: KaryoFlow uses ResNet-50–FPN, whereas RTMDet-L uses CSPNeXt-L and DINO uses multi-scale deformable attention [51], [3]. We therefore describe these as best results within the checkpoint-linked comparison, rather than claiming an exhaustive cross-study SOTA.

a) Small-object performance: Dataset 1 is the small-object-heavy cohort. LQCR reaches validation $AP_S = 0.5187$, close to Cascade R-CNN (0.5200) and above DINO R50 (0.4920) and RTMDet-L (0.5040). On its held-out test split, the three-seed LQCR mean is 0.7480 ± 0.0017 mAP and 0.5147 ± 0.0090 AP_S, improving the uncalibrated KaryoFlow means by 0.0067 and 0.0057. On Dataset 2, LQCR leads overall mAP but remains below DINO on AP_S. This pattern supports the intended role of quality calibration—better aggregate ordering across strict IoU thresholds—without claiming universal superiority at every scale.

b) Class- and scale-resolved analysis: The diagnostic exports show that residual errors are structured rather than uniform. High-IoU recall falls most for small chromosomes and locally overlapping instances (Figure 3), matching the training-set statistics in Section IV-A. At the conventional IoU 0.50 threshold, performance is already near saturation; the remaining headroom therefore lies in sub-pixel coordinate refinement and quality-aware ordering, not simply in producing more low-quality proposals. This observation motivates LQCR and explains why its gain concentrates at AP90/AP95.

c) Causal interpretation of controlled comparisons: The same-checkpoint solver comparison is accuracy-neutral, and standardized multi-seed coupling experiments do not establish an accuracy contribution. The retained positive intervention is LQCR: enabling its score only at the final output changes mAP from 0.8630 to 0.8704 on exactly the same checkpoint and detections before ranking. This paired construction isolates decision-stage calibration from generation, interaction, and numerical integration effects. We report this as a fixed-checkpoint causal intervention, not as a cross-seed training gain; the independent multi-seed evidence in the paper concerns RF, prediction parameterization, renewal, and coupling.

The monotone mAP trend supports using $\beta = 2$, while invariant AP50 shows that calibration reorders already detected candidates rather than creating new easy true positives. AP90 and AP95 rise at every point in the sweep, matching the threshold-factorization prediction.

1) Scale-Resolved Cross-Dataset Comparison: Figure 5 replaces selected visual cases with complete-set evidence. Dataset 1 has a substantially smaller relative-box distribution

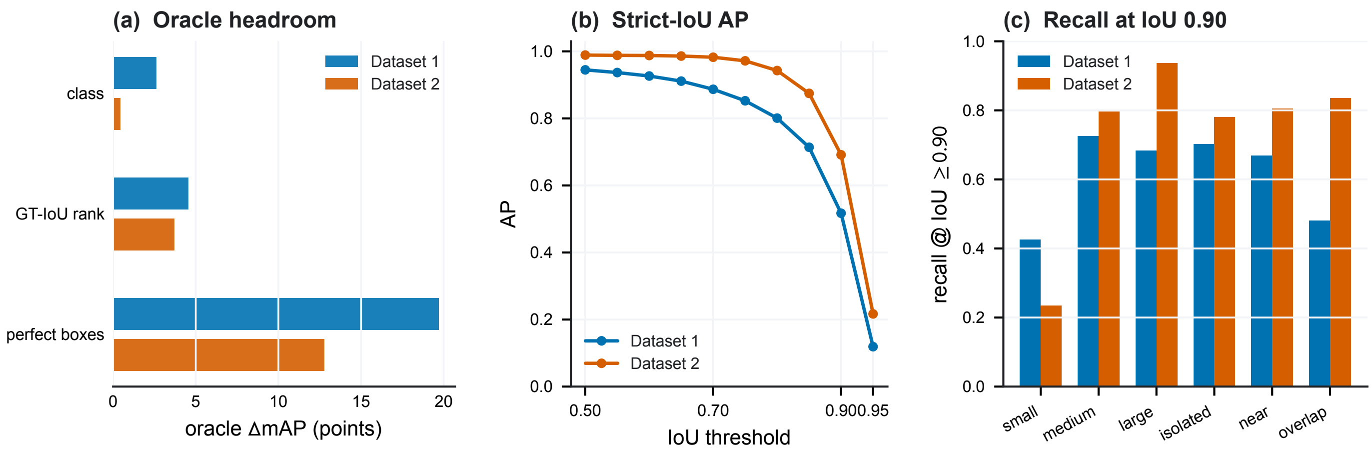

Fig. 3. Precision bottleneck diagnosis. (a) Oracle mAP headroom for perfect classes, ground-truth-IoU ranking, and perfect coordinates. Ground-truth-IoU ranking is a diagnostic upper bound, not a trained LQCR result. (b) AP versus IoU threshold. (c) Detection recall at IoU 0.9000 by scale and overlap condition. Dataset 1 uses a seed-42 100-image diagnostic subset; Dataset 2 uses the seed-42 500-image validation export.

TABLE IV
EVIDENCE-CONTROLLED ABLATION ON DATASET 2 VALIDATION. THE FINAL TWO ROWS SHARE THE SEED-42 RF-DPM++ CHECKPOINT; LQCR CHANGES ONLY OUTPUT RANKING.

Experiment	Solver	Steps	NFE	mAP	AP50	AP75	AP_S	AP_M	AP_L
DDPM baseline (Euler 1-step)	Euler	1	1	0.7740	0.9680	0.9160	0.3170	0.7730	0.7750
RF+Heun (KaryoFlow)	Heun	4	7	**0.8560**	0.9890	0.9690	0.5020	0.8530	0.8670
DPM-Solver++ (seed 42)	DPM++	4	4	0.8630	0.9889	0.9717	0.5315	0.8594	0.8921
+LQCR (final-only)	DPM++	4	4	**0.8704**	0.9888	0.9722	0.5304	0.8666	0.9142

TABLE V
CONTROLLED METHOD AND EFFICIENCY ANALYSES. MEANS USE THREE SEEDS UNLESS MARKED SEED 42. CHE DENOTES CASCADE-HEAD EVALUATIONS AND IS DISTINCT FROM SOLVER NFE.

Intervention	Controlled comparison	Accuracy effect	Efficiency effect	Interpretation / boundary
LQCR	D1, 3 seeds / D2, seed 42	+0.0080/ +0.0074 mAP	+3.3% latency	Final ranking; candidate states fixed before post-processing
Renewal off	D2 / D1, $K = 500$	−0.0003/ −0.0030	−2.4% latency	Restores DPM++ history identity; fails at $K = 100$ (−0.0313)
Head distillation	D2, H6 teacher vs H3 student	−0.0040 mAP	1.71×; CHE 24 → 12	NFE remains four; no D1 distillation claim
GACS	D1, adaptive vs fixed two-step	−0.0007 mAP	−13.8% latency	Not lossless versus four-step reference (0.7470)
x_0 prediction	D1 / D2, 3-seed vs velocity	+0.0020/ +0.0020 mAP	no added cost	Matches t^{-2} conditioning analysis; test difference is noise-scale

than Dataset 2 (median 0.4% versus 0.8%). On the Dataset 1 test split, KaryoFlow+LQCR reaches $AP_S = 0.5147 \pm 0.0090$ over three seeds, a 0.0057 gain over the uncalibrated model and 0.0147 above the strongest listed comparator. On Dataset 2 validation, LQCR raises AP_S to 0.5304 but remains below DINO R50 (0.5530). The result supports scale-dependent small-object competitiveness rather than a universal advantage. Because COCO's small-area threshold is absolute, raw AP_S levels are interpreted only within each dataset; the relative-area CDF is used to characterize the cross-dataset scale shift.

E. Controlled Coupling Result

With standardized augmentation, the three coupling strategies are equivalent within cross-seed variation on both datasets. On Dataset 1, their three-seed means lie within approximately 0.0020 mAP; on Dataset 2, the stochastic-minus-random difference is also non-significant. Earlier larger gains pooled incompatible experimental lineages and are superseded. We retain this result to document the falsified direction, while excluding Stochastic Coupling from the model definition and contribution list.

F. Solver Analysis

a) DPM-Solver++ step ablation: DPM-Solver++ converges at 2 steps (mAP 0.8630, seed 42); no benefit beyond 2 steps confirms RF trajectories are near-straight.

b) DPM-Solver++ vs Heun at matched NFE: At similar NFE, DPM-Solver++ 4-step (4 NFE, 0.8630) ≈ Heun 2-step (3 NFE, 0.8630) — equal accuracy, so DPM-Solver++ achieves

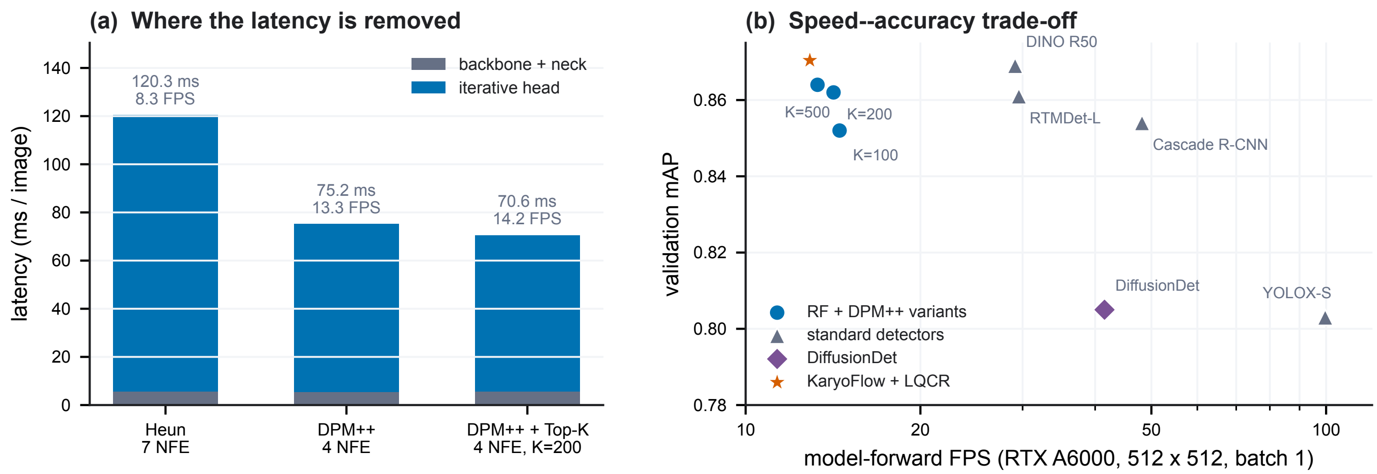

Fig. 4. Same-hardware efficiency on Dataset 2. (a) Latency decomposition. (b) Validation mAP versus FPS on one RTX A6000. The three blue KaryoFlow operating points use $K = 500$, 200, and 100 without connecting lines; the highlighted LQCR point is the complete accuracy model.

TABLE VI

VALIDATION COMPARISON ON TWO CHROMOSOME DATASETS. DATASET 1 KARYOFLOW+LQCR IS THE CLEAN THREE-SEED MEAN; DATASET 2 IS THE CONTROLLED FINAL-ONLY SEED-42 MODEL. OTHER ROWS ARE CHECKPOINT-LINKED RE-EVALUATIONS. BOLD DENOTES THE BEST MAP WITHIN THIS COMPARISON, NOT AN EXHAUSTIVE CROSS-PAPER SOTA CLAIM.

		Dataset 1				Dataset 2			
Method	Backbone	mAP	AP50	AP75	AP_S	mAP	AP50	AP75	AP_S
KaryoFlow+LQCR	ResNet-50	**0.7550**	0.9413	0.8420	0.5187	**0.8704**	0.9888	0.9722	0.5304
DINO R50	ResNet-50	0.7420	0.9400	0.8310	0.4920	0.8690	0.9910	0.9770	0.5530
RTMDet-L	CSPNeXt-L	0.7420	0.9460	0.8540	0.5040	0.8610	0.9910	0.9720	0.5220
Cascade R-CNN	ResNet-50	0.7320	0.9320	0.8430	0.5200	0.8540	0.9860	0.9660	0.4140
DiffusionDet	ResNet-50	0.6380	0.8710	0.7440	0.3550	0.8050	0.9710	0.9370	0.4030
YOLOX-S	CSPDarkNet-S	0.5890	0.9240	0.7040	0.3590	0.8030	0.9860	0.9460	0.4410

TABLE VII

LEARNED-QUALITY EXPONENT SWEEP ON DATASET 2 VALIDATION (SEED 42). THE $\beta = 0$ ROW DISABLES QUALITY RANKING. THESE ARE LEARNED-q RESULTS, NOT GROUND-TRUTH-IOU ORACLE RANKINGS; ALL ROWS USE THE SAME SINGLE-IMAGE PRECISION-DIAGNOSTIC PROTOCOL.

β	mAP	ΔmAP	AP50	AP75	AP90	AP95
0	0.8630	—	0.9889	0.9717	0.6914	0.2167
0.2500	0.8657	+0.0027	0.9889	0.9718	0.7022	0.2284
0.5000	0.8673	+0.0043	0.9889	0.9719	0.7086	0.2352
1	0.8689	+0.0059	0.9888	0.9717	0.7158	0.2422
2	**0.8704**	**+0.0074**	0.9888	0.9722	**0.7218**	**0.2492**

42.9% fewer NFE at no cost. At matched *step count* on the same RF-DPM++ checkpoint, DPM-Solver++ gives 0.8630 and Heun gives 0.8640 ($\Delta = -0.0010$), which is practically neutral. We therefore attribute no accuracy gain to the off-the-shelf solver; its value is the lower evaluation count and latency.

c) Test set evaluation: On the Dataset 2 test set, DPM-Solver++ achieves mAP 0.8590 (vs val 0.8630 for seed 42, $\Delta = -0.0040$), confirming that the aggregate accuracy generalizes well. The AP_S point estimate, however, swings from 0.4990 (val) to 0.5770 (test) for the same checkpoint, indicating that AP_S on this 24-class benchmark is high-variance (only 60 val images contain small objects). We therefore base the small-object discussion on the full scale- and overlap-resolved diagnostics rather than on one split-specific AP_S point estimate.

G. FPS / Latency Benchmark

Table VIII and Figure 4 report the speed-accuracy trade-off at 512×512 input resolution. Latency is CUDA-event model-forward time and excludes image decoding, host pre-processing, and visualization. The unpruned $K = 500$ point and the $K = 200/100$ pruning points share one checkpoint and evaluation protocol. DPM-Solver++ with Top-K pruning reaches a latency compatible with interactive use, while single-shot detectors (Cascade R-CNN, YOLOX-S) trade accuracy for speed.

DPM-Solver++ +Top-K (K=200) reaches 70.57 ms / 14.2 FPS at mAP 0.8620. The complete KaryoFlow+LQCR model reaches 77.68 ms / 12.9 FPS at mAP 0.8704; its 2.48 ms (3.3%) overhead over uncalibrated DPM-Solver++ buys the highest accuracy in the comparison. The cascade head dominates 90%+ of latency; the backbone+neck is a minor cost ($\sim$ 5.6 ms). These speed-linked re-evaluations differ by at most 0.0030 from the canonical training-time point estimates in Tables IV and VI, which is consistent with random proposal initialization. The RF+Heun row uses the same Dataset 2 A1 checkpoint for both axes.

The efficiency analysis separates three counts that are often conflated: DPM-Solver++ reduces solver NFE ($7 \rightarrow 4$), head

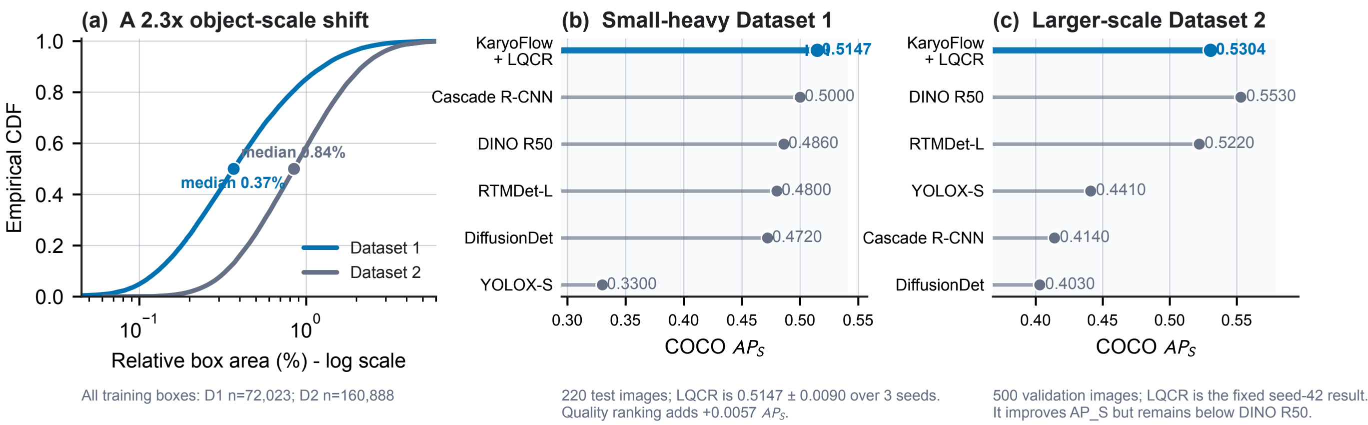

Fig. 5. Object-scale shift and small-object detection across two chromosome cohorts. (a) Empirical CDF of relative ground-truth box area using all training annotations. Dataset 1 boxes are markedly smaller than Dataset 2 boxes. (b) Dataset 1 test AP$_S$ on 220 images; KaryoFlow+LQCR is the clean three-seed mean. (c) Dataset 2 validation AP$_S$ on 500 images; LQCR is the controlled seed-42 final-only result and other detectors use unified checkpoint-linked re-evaluation. AP$_S$ is interpreted within, not across, datasets.

TABLE VIII
SAME-HARDWARE MODEL-FORWARD LATENCY BENCHMARK ON DATASET 2 AT 512×512 RESOLUTION (RTX A6000, BATCH SIZE 1, 10 WARM-UP AND 500 TIMED CUDA-EVENT ITERATIONS). ACCURACY IS RE-EVALUATED ON VALIDATION WITH THE SPEED-LINKED CHECKPOINT.

Model	Solver	NFE	Latency (ms)	FPS	mAP
RF+Heun	Heun	7	120.34	8.3	0.8560
DPM-Solver++	DPM++	4	**75.20**	**13.3**	**0.8640**
KaryoFlow+LQCR	DPM++	4	**77.68**	**12.9**	**0.8704**
DPM-Solver++ +Top-K (K=200)	DPM++	4	**70.57**	**14.2**	**0.8620**
DPM-Solver++ +Top-K (K=100)	DPM++	4	68.90	14.5	0.8520
DINO R50	—	1	34.33	29.1	0.8690
RTMDet-L	—	1	33.83	29.6	0.8610
Cascade R-CNN	—	1	20.76	48.2	0.8540
DiffusionDet	Euler	1	24.09	41.5	0.8050
YOLOX-S	—	1	10.04	99.7	0.8030

distillation reduces CHE ($24 \rightarrow 12$ at four NFE), and GACS reduces mean per-image steps ($2 \rightarrow 1.6136$). Renewal removal does not change any count; it makes multistep history coherent and eliminates proposal replacement overhead.

H. Sensitivity to Annotation Precision

To quantify why strict localization matters, we perturb only the Dataset 2 test annotations while keeping images and predictions fixed. A 5-pixel Gaussian center perturbation reduces AP50 by 0.1410 but AP75 by 0.5960. This is an evaluation-sensitivity diagnostic rather than a model-robustness claim: modest annotation uncertainty is amplified at strict IoU, further motivating AP90/AP95 analysis and localization-aware ranking for small chromosomes.

V. DISCUSSION

KaryoFlow's components address different consequences of metaphase-image structure. The low-dimensional linear path and direct box prediction reduce optimization burden when annotated cohorts are modest. Proposal identity matters because dozens of adjacent candidates evolve simultaneously, while the nearly fixed chromosome count permits a redundant proposal pool that makes renewal unnecessary at the retained operating point. LQCR then addresses the remaining decision bottleneck: class confidence is already sufficient at IoU 0.5000, but it does not identify which small or overlapping boxes will survive stricter localization thresholds. The concentration of LQCR gains at AP90 and AP95 is consistent with this task-derived mechanism.

The efficiency results reveal complementary deployment controls rather than a single opaque speedup. DPM-Solver++ reduces solver evaluations, distillation reduces cascade-head evaluations, Top-K reduces proposals after the first step, and GACS reduces mean per-image steps. This decomposition enables an accuracy-oriented setting (0.8704 mAP, 77.68 ms) or a compressed setting (0.8590 mAP, 44.10 ms) without changing the central trajectory–ranking formulation.

The study has three boundaries. First, both cohorts are public chromosome datasets and lack patient identifiers, so patient-level leakage and clinical workflow outcomes cannot be assessed. Second, LQCR is replicated across three clean Dataset 1 training seeds and a held-out test split, but Dataset 2 uses one fixed validation checkpoint and the same cohort selects β. A locked Dataset 2 test and natural-image validation are still required to establish broader generality. Third, history-consistent inference relies on proposal redundancy: the observed $K = 100$ failure should be treated as an operating constraint, not hidden by the high-K result. These limits do not affect the box-path derivations, but they define the present empirical scope.

VI. CONCLUSION

KaryoFlow aligns trajectory learning, multistep state identity, and final localization-aware ranking for dense chromosome detection. RF supplies the few-step generative foundation; direct data prediction improves three-seed mAP by 0.0020 on both datasets; and the proposal-identity analysis makes DPM++ history mathematically coherent at the retained proposal count. The independent decision contribution, LQCR, improves validation mAP by 0.0080 across three Dataset 1 seeds and by 0.0074 in the Dataset 2 fixed-checkpoint intervention. On Dataset 2, AP90/AP95 improve by $0.0304/0.0325$

without changing candidate boxes before ranking. Together with orthogonal inference compression, these results establish a competitive and interpretable generative detector for chromosome karyotyping.

REFERENCES

[1] Z. Cai and N. Vasconcelos, “Cascade R-CNN: Delving into high quality object detection,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 6154–6162.

[2] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, “YOLOX: Exceeding YOLO series in 2021,” arXiv:2107.08430, 2021. [Online]. Available: https://arxiv.org/abs/2107.08430

[3] H. Zhang, F. Li, S. Liu, L. Zhang, H. Su, J. Zhu, L. M. Ni, and H.-Y. Shum, “DINO: DETR with improved denoising anchor boxes for end-to-end object detection,” in *Int. Conf. Learn. Represent. (ICLR)*, 2023.

[4] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, “Deformable DETR: Deformable transformers for end-to-end object detection,” in *Int. Conf. Learn. Represent. (ICLR)*, 2021.

[5] S. Chen, P. Sun, Y. Song, and P. Luo, “DiffusionDet: Diffusion model for object detection,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 1980–1989.

[6] X. Liu, C. Gong, and Q. Liu, “Flow straight and fast: Learning to generate and transfer data with rectified flow,” in *Int. Conf. Learn. Represent. (ICLR)*, 2023.

[7] C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, and J. Zhu, “DPM-Solver++: Fast solver for guided sampling of diffusion probabilistic models,” *Machine Intelligence Research*, 2025.

[8] W. C. Schwartzkopf, A. C. Bovik, and B. L. Evans, “Maximum-likelihood techniques for joint segmentation–classification of multispectral chromosome images,” *IEEE Transactions on Medical Imaging*, vol. 24, no. 12, pp. 1593–1610, 2005.

[9] Y. Qin, J. Wen, H. Zheng, X. Huang, J. Yang, N. Song, Y.-M. Zhu, L. Wu, and G.-Z. Yang, “Varifocal-Net: A chromosome classification approach using deep convolutional networks,” *IEEE Transactions on Medical Imaging*, vol. 38, no. 11, pp. 2569–2581, 2019.

[10] L. Xiao, C. Luo, T. Yu, Y. Luo, M. Wang, F. Yu, Y. Li, C. Tian, and J. Qiao, “DeepACEv2: Automated chromosome enumeration in metaphase cell images using deep convolutional neural networks,” *IEEE Transactions on Medical Imaging*, vol. 39, no. 12, pp. 3920–3932, 2020.

[11] J. Li, S. Zheng, Z. Shui, S. Zhang, L. Yang, Y. Sun, Y. Zhang, H. Li, Y. Ye, P. M. A. van Ooijen, K. Li, and L. Yang, “Masked conditional variational autoencoders for chromosome straightening,” *IEEE Transactions on Medical Imaging*, vol. 43, no. 1, pp. 216–228, 2024.

[12] C.-E. Kuo, J.-Z. Li, J.-J. Tseng, F.-C. Lo, M.-J. Chen, and C.-H. Lu, “ChromosomeNet: Deep learning-based automated chromosome detection in metaphase cell images,” *IEEE Open Journal of Engineering in Medicine and Biology*, vol. 6, pp. 227–236, 2024.

[13] S. Matsumoto, M. Sojo, K. Oshima, K. Nishikawa, K. Okubo *et al.*, “Detection and classification of chromosomes with sister chromatid cohesion defects using object detection models,” *Scientific Reports*, vol. 16, p. 13719, 2026.

[14] C. Rosas-Alatriste, N. O. Rodríguez-Rodríguez, A. J. Argüelles-Cruz, A. Rosas-Cabral, C. Yáñez-Márquez *et al.*, “Automated karyotyping and structural anomaly detection through a hybrid multi-stage deep learning framework integrating chromosome detection, pairwise classification, and autoencoder-based analysis,” *Scientific Reports*, vol. 16, p. 22469, 2026.

[15] T. Chen, C. Xie, W. Zhang *et al.*, “ImageNet pre-training and two-step transfer learning in chromosome image classification,” *Scientific Reports*, vol. 16, p. 7572, 2026.

[16] C. Yang *et al.*, “A comprehensive foundation model for generalizable cytogenetics in precision oncology with CHROMA,” *npj Precision Oncology*, vol. 10, p. 187, 2026.

[17] E. Baty, C. P. Bridges, and S. Hadfield, “FlowDet: Unifying object detection and generative transport flows,” arXiv:2512.16771, 2025. [Online]. Available: https://arxiv.org/abs/2512.16771

[18] H. Lee, C. Lee, N. Seo, J. S. Lim, and H. Hong, “DeFloMat: Detection with flow matching for stable and efficient generative object localization,” arXiv:2512.22406, 2025. [Online]. Available: https://arxiv.org/abs/2512.22406

[19] X. Chen, Z. Liu, K. Z. Luo, S. Datta, A. Polavaram, Y. Wang *et al.*, “DiffuBox: Refining 3D object detection with point diffusion,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2024.

[20] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, “Flow matching for generative modeling,” in *Int. Conf. Learn. Represent. (ICLR)*, 2023.

[21] C. Jones, E. Noutahi, J. Hartford, and C. Eastwood, “Elucidating the design space of flow matching for cellular microscopy,” arXiv:2603.26790, 2026. [Online]. Available: https://arxiv.org/abs/2603.26790

[22] F. Nützel, M. Dombrowski, and B. Kainz, “Flow matching with optimized subclass priors for medical image augmentation,” arXiv:2605.16469, 2026. [Online]. Available: https://arxiv.org/abs/2605.16469

[23] M. Tivnan, J. Teneggi, T.-C. Lee, R. Zhang, K. Boedeker, L. Cai, G. J. Gang, J. Sulam, and J. W. Stayman, “Fourier diffusion models: A method to control MTF and NPS in score-based stochastic image generation,” *IEEE Transactions on Medical Imaging*, vol. 44, no. 9, pp. 3694–3704, 2025.

[24] H. Yang, W. Hua, Z. Xu, and J. Sun, “Domain-generalized discrete diffusion model for cross-domain medical image segmentation,” *IEEE Transactions on Medical Imaging*, vol. 44, no. 11, pp. 4131–4142, 2025.

[25] X. Shen, H. Huang, B. Nichyporuk, and T. Arbel, “Improving robustness and reliability in medical image classification with latent-guided diffusion and nested-ensembles,” *IEEE Transactions on Medical Imaging*, vol. 44, no. 12, pp. 4890–4902, 2025.

[26] W. Yang, S. Gao, H. Zhang, H. Yu, M. Xu, P. Chong, W. Zhang, H. Wang, W. Zhang, and A. Qian, “PtbNet: Based on local few-shot classes and small objects to accurately detect PTB,” *IEEE Transactions on Medical Imaging*, vol. 43, no. 12, pp. 4331–4343, 2024.

[27] K. Yan, J. Cai, Y. Zheng, A. P. Harrison, D. Jin, Y. Tang, Y. Tang, L. Huang, J. Xiao, and L. Lu, “Learning from multiple datasets with heterogeneous and partial labels for universal lesion detection in CT,” *IEEE Transactions on Medical Imaging*, vol. 40, no. 10, pp. 2759–2770, 2021.

[28] C. Chen, J. Wang, J. Pan, C. Bian, and Z. Zhang, “GraphSKT: Graph-guided structured knowledge transfer for domain adaptive lesion detection,” *IEEE Transactions on Medical Imaging*, vol. 42, no. 2, pp. 507–518, 2023.

[29] Y. Ren, X. Liu, J. Ge, Z. Liang, X. Xu, L. J. Grimm, J. Go, J. R. Marks, and J. Y. Lo, “Ipsilateral lesion detection refinement for tomosynthesis,” *IEEE Transactions on Medical Imaging*, vol. 42, no. 10, pp. 3080–3090, 2023.

[30] R. Wang, B. Chen, D. Meng, and L. Wang, “Weakly supervised lesion detection from fundus images,” *IEEE Transactions on Medical Imaging*, vol. 38, no. 6, pp. 1501–1512, 2019.

[31] S.-H. Bae and K.-J. Yoon, “Polyp detection via imbalanced learning and discriminative feature learning,” *IEEE Transactions on Medical Imaging*, vol. 34, no. 11, pp. 2379–2393, 2015.

[32] B. Jiang, R. Luo, J. Mao, T. Xiao, and Y. Jiang, “Acquisition of localization confidence for accurate object detection,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 784–799.

[33] X. Li, W. Wang, L. Wu, S. Chen, X. Hu, J. Li, J. Tang, and J. Yang, “Generalized focal loss: Learning qualified and distributed bounding boxes for dense object detection,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 21 002–21 012.

[34] H. Zhang, Y. Wang, F. Dayoub, and N. Sunderhauf, “VarifocalNet: An IoU-aware dense object detector,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 8514–8523.

[35] A. Mehrtash, W. M. Wells, C. M. Tempany, P. Abolmaesumi, and T. Kapur, “Confidence calibration and predictive uncertainty estimation for deep medical image segmentation,” *IEEE Transactions on Medical Imaging*, vol. 39, no. 12, pp. 3868–3878, 2020.

[36] P. Dhariwal and A. Nichol, “Diffusion models beat GANs on image synthesis,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021.

[37] A. Tong, K. Fatras, N. Malkin, G. Huguet, Y. Zhang, J. Rector-Brooks, G. Wolf, and Y. Bengio, “Conditional flow matching: Simulation-free dynamic optimal transport,” in *Int. Conf. Learn. Represent. (ICLR)*, 2023.

[38] A.-A. Pooladian, H. Ben-Hamu, C. Domingo-Enrich, B. Amos, Y. Lipman, and R. T. Q. Chen, “Multisample flow matching: Straightening flows with minibatch couplings,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2023, pp. 28 100–28 127.

[39] M. Cuturi, “Sinkhorn distances: Lightspeed computation of optimal transport,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2013, pp. 2292–2300.

[40] H. K. Cheng and A. Schwing, “The curse of conditions: Analyzing and improving optimal transport for conditional flow-based generation,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 15 875–15 884.

[41] South China Normal University, "RST (chromosome20240904) dataset," Roboflow Universe, 2024, accessed Jul. 2026. [Online]. Available: https://universe.roboflow.com/south-china-normal-university-imqzk/rst

[42] J.-J. Tseng, C.-H. Lu, J.-Z. Li, H.-Y. Lai, M.-H. Chen, F.-Y. Cheng, and C.-E. Kuo, "An open dataset of annotated metaphase cell images for chromosome identification," *Scientific Data*, vol. 10, p. 104, 2023.

[43] C.-H. Lu, C.-E. Kuo, and J.-J. Tseng, "CIL:54816 - 24 chromosomes object dataset," Cell Image Library, 2022. [Online]. Available: https://doi.org/10.7295/W9CIL54816

[44] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.

[45] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 936–944.

[46] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Int. Conf. Learn. Represent. (ICLR)*, 2019.

[47] ——, "SGDR: Stochastic gradient descent with warm restarts," in *Int. Conf. Learn. Represent. (ICLR)*, 2017.

[48] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 2999–3007.

[49] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 658–666.

[50] H. W. Kuhn, "The hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1-2, pp. 83–97, 1955.

[51] C. Lyu, W. Zhang, H. Huang, Y. Zhou, Y. Wang, Y. Liu, S. Zhang, and K. Chen, "RTMDet: An empirical study of designing real-time object detectors," arXiv:2212.07784, 2022.